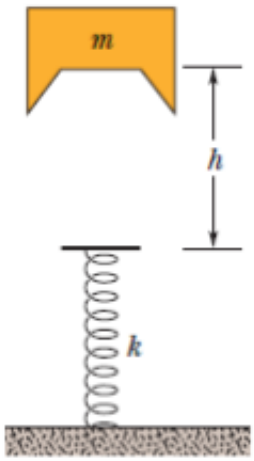
A 3.00×10^3 N automobile accelerates from rest to 50.0 m/s in 6.00 s with a constant acceleration. Calculate the instantaneous power delivered by the engine at t = 6.00 s.

A) 12.8×10^4 W
B) 15.0×10^4 W
C) 9.45×10^5 W
D) 24.5×10^3 W
E) 4.51×10^7 W

A 5.0 kg box is taken from point A (1.0, 2.0, – 2.0) m to point B (6.0, – 3.0, – 2.0) m by applying a single constant force **F** = (3.0 N) **i** + (2.0 N) **j** + (5.0 N) **k**. Find the change in the kinetic energy of the box.

A) 5.0 J
B) 25 J
C) 10 J
D) 18 J
E) 2.0 J

A massless spring has a spring constant of 500 N/m. A 2.0 kg object is released from rest at a height h = 1.0 m above the spring and lands on it (**Figure**). Find the object's speed when the spring is compressed 20 cm.

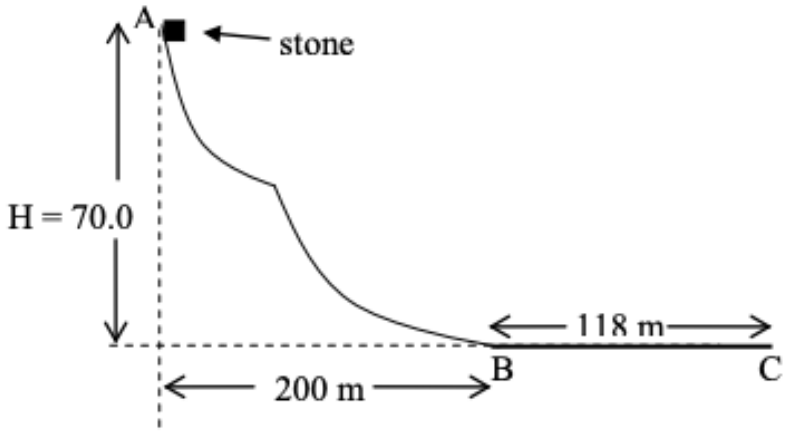


A) 3.7 m/s
B) 3.1 m/s
C) 4.1 m/s
D) 4.5 m/s
E) 4.9 m/s

Two particles of masses 3.0 kg and 5.0 kg are moving with initial velocities of (–3.0 **i** + 4.0 **j**) m/s and (2.0 **i** + 3.0 **j**) m/s respectively. They collide completely inelastically. Find the velocity of the center of mass of the two particles after the collision.

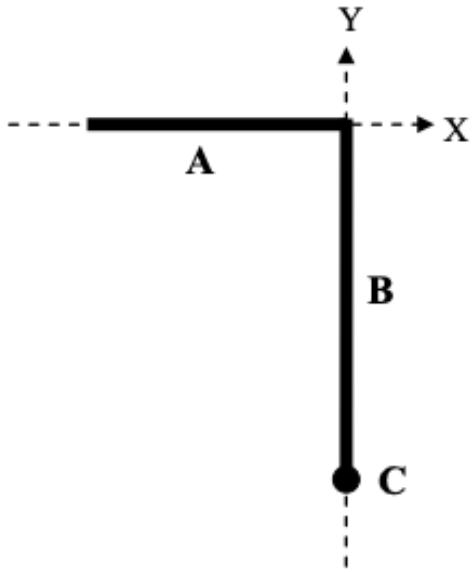
A) (0.13 **i** + 3.4 **j**) m/s
B) (3.2 **i** + 4.4 **j**) m/s
C) (5.13 **i** + 1.34 **j**) m/s
D) (–9.00 **i** + 12.0 **j**) m/s
E) (10.0 **i** + 15.0 **j**) m/s

A 15.0 kg stone slides down a smooth snow-covered hill (**Figure**), leaving point A with a speed of 4.0 m/s. Then it slides a distance of 118 m on a rough horizontal surface from point B to point C before coming to rest. Find the coefficient of kinetic friction μ_k between the stone and the surface.



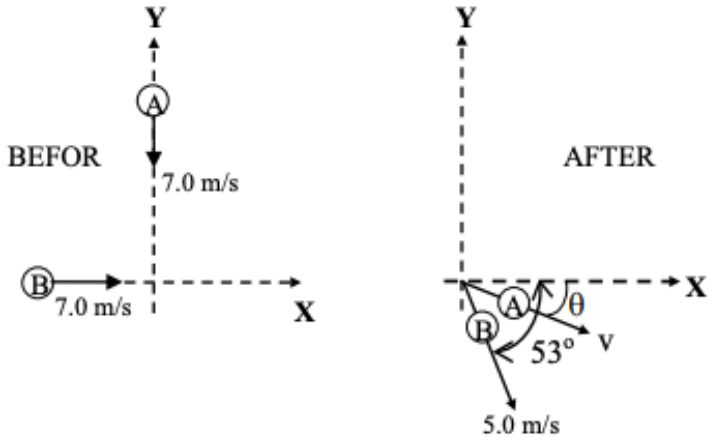
A) 0.600
B) 0.500
C) 0.400
D) 0.550
E) 0.450

A machine part consists of three objects welded together: A) a thin, uniform 4.00 kg bar that is 1.50 m long, B) a vertical bar of mass 3.00 kg and length 1.80 m and C) dense 2.00 kg ball attached to the end of object B (**Figure**). Find the center of mass of this system.



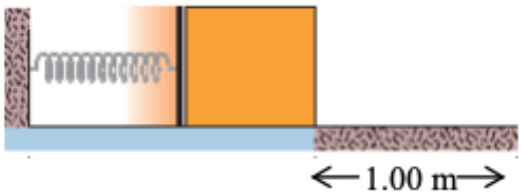
A) (– 0.333m, – 0.700 m)
B) (– 0.750m, – 0.900 m)
C) (– 0.250m, – 0.450 m)
D) (– 0.453 m, – 0.767 m)
E) (– 0.670 m, – 0.767 m)

Two objects A and B, with the same mass collide on ice with negligible friction. **Figure** gives speeds and directions of the objects BEFORE and AFTER the collision. Find the speed v and angle θ for object A after the collision.



A) 5.0 m/s , 37°
B) 7.0 m/s , 45°
C) 10 m/s, 30°
D) 3.5 m/s, 50°
E) 1.4 m/s, 20°

A block with mass m = 0.50 kg is forced against a horizontal spring of spring constant 100 N/m and negligible mass, compressing the spring a distance of 0.20 m (**Figure**). When released, the block moves on a horizontal tabletop for 1.0 m before coming to rest. Find the coefficient of kinetic friction μ_k between the block and the tabletop.



A) 0.41
B) 0.35
C) 0.25
D) 0.13
E) 0.53

A car with a mass of 1.2 x10^3 kg is travelling to the right at a speed of 15 m/s when it collides head-on with a truck of mass 2.0 x10^3 kg travelling at a speed of 15 m/s to the left. The vehicles lock together when they collide. Find the average force (both magnitude and direction) exerted on the car if the collision lasts for 0.20 s.

A) 1.1 x 10^5 N to the left
B) 1.1 x 10^5 N to the right
C) 2.2 x 10^4 N to the left
D) 3.1 x 10^4 N to the right
E) 5.3 x 10^5 N to the left